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Safety data sheet

BASF Safety data sheet according to Regulation UK SI 2019/758 and UK SI 2020/1577 as amended from time to time.
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Product: **Loxanol® MI 6311**
(ID no. 30756558/SDS_GEN_GB/EN)
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SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Loxanol® MI 6311

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: additives for inks, varnishes or coatings, stabilizer, Paint additive
Recommended use: Chemical, for industrial use only

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF plc
4th and 5th Floors, 2 Stockport Exchange
Railway Road, Stockport, SK1 3GG
UNITED KINGDOM

Telephone: +44 161 475 3000
E-mail address: product-safety-uk-and-ireland@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

No need for classification according to GHS criteria for this product.

2.2. Label elements

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

The product does not require a hazard warning label in accordance with GHS criteria.

2.3. Other hazards

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

imidazole derivative, in water

Hazardous ingredients (GHS)

No particular hazards known.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Rinse mouth and then drink 200-300 ml of water.

4.2. Most important symptoms and effects, both acute and delayed
Symptoms: (Further) symptoms and / or effects are not known so far

Hazards: (Further) symptoms and / or effects are not known so far

4.3. Indication of any immediate medical attention and special treatment needed
Treatment: Symptomatic treatment (decontamination, vital functions).

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media
Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

5.2. Special hazards arising from the substance or mixture
Endangering substances: nitrogen oxides, carbon oxides
Advice: The substances/groups of substances mentioned can be released in case of fire. Under certain conditions in case of fire other hazardous combustion products may be generated.

5.3. Advice for fire-fighters
Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures
Handle in accordance with good industrial hygiene and safety practice.

6.2. Environmental precautions
Discharge into the environment must be avoided.

6.3. Methods and material for containment and cleaning up
For small amounts: Pick up with suitable appliance and dispose of.
For large amounts: Pick up with suitable appliance and dispose of.
Incinerate or take to a special waste disposal site in accordance with local authority regulations.

6.4. Reference to other sections
Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
Keep away from sources of ignition - No smoking.

7.2. Conditions for safe storage, including any incompatibilities

Segregate from acids and acid forming substances.
Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

Storage stability:
Storage temperature: <= 40 °C
Storage duration: 12 Months
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

No substance specific occupational exposure limits known.

Components with PNEC

3699-54-5: 1-(2-Hydroxyethyl)imidazolidin-2-one
freshwater: 0.1 mg/l
marine water: 0.01 mg/l
intermittent release: 1 mg/l
STP: 112.3 mg/l
sediment (freshwater): 0.37 mg/kg
sediment (marine water): 0.037 mg/kg
soil: 0.0153 mg/kg

Components with DNEL

3699-54-5: 1-(2-Hydroxyethyl)imidazolidin-2-one
worker: Long-term exposure- systemic effects, Inhalation: 44.1 mg/m3
worker: Long-term exposure- systemic effects, dermal: 125 mg/kg
consumer: Long-term exposure- systemic effects, Inhalation: 10.9 mg/m3
consumer: Long-term exposure- systemic effects, dermal: 62.5 mg/kg
consumer: Long-term exposure- systemic effects, oral: 6.3 mg/kg

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection required in case of exceptional circumstances (e.g.: accidental release, exceeding the occupational exposure limit) Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
butyl rubber (butyl) - 0.7 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
In order to satisfy general industrial hygiene rules safety glasses with side-shields (e.g. EN 166) are recommended.

Body protection:
No body protection required if used for intended purpose and satisfying generally accepted industrial hygiene rules.

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. Store work clothing separately. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

Form:	liquid
Colour:	colourless to yellowish
Odour:	amine-like
Odour threshold:	Not determined due to potential health hazard by inhalation.
pH value:	9 - 11 (100 g/l, 20 °C) Information based on the main component/s.
Melting point:	< -30 °C

decomposition point:	> 150 °C (996 hPa) Information based on the main component/s., Product turns dark. The substance / product decomposes.	(OECD Guideline 103)
Flash point:	No flash point - Measurement made up to pilot light extinguishes.	(ISO 2719, closed cup)
	No flash point - Measurement made up to pilot light extinguishes.	(ISO 13736, closed cup)
	No flash point - Measurement made up to pilot light extinguishes.	(ISO 2592, open cup)
Lower explosion limit:	For liquids not relevant for classification and labelling.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one		
Ignition temperature:	*375 °C*	*(DIN 51794)*

Vapour pressure:	0.000013 hPa (25 °C) Information based on the main component/s.	(OECD Guideline 104)
Density:	1.311 g/cm3 (20 °C, 1,013 hPa) Information based on the main component/s.	(OECD Guideline 109)
Relative vapour density (air):	Water content greater than 10%.	
Solubility in water:	Information based on the main component/s. 982 g/l (20 °C, 1,013 hPa, pH 10.4)	(OECD Guideline 105)
Partitioning coefficient n-octanol/water (log Kow):	-0.7 (20 °C; pH value: 10) Information based on the main component/s.	(OECD Guideline 107)
Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one		
Thermal decomposition:	*380 °C, 140 kJ/kg, (DSC (OECD 113))*	

Viscosity, dynamic:	27.4 mPa.s (25 °C)	(ASTM D 7042)
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	

9.2. Other information

SADT: Study scientifically not justified.
Miscibility with water:
completely (e.g. >=90%)
Hygroscopy: Non-hygroscopic
Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Adsorption/water - soil: log KOC: -0.096 *(calculated)*

Grain size distribution: The substance / product is marketed or used in a non solid or granular form.

SECTION 10: Stability and Reactivity

10.1. Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated.

10.2. Chemical stability
The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

10.4. Conditions to avoid
Avoid extreme heat.

10.5. Incompatible materials

Substances to avoid:
strong bases, strong acids

10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on toxicological effects

Acute toxicity

Experimental/calculated data:
(by inhalation):No data available.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one

Experimental/calculated data:
LD50 rat (oral): > 2,000 mg/kg (similar to OECD guideline 401)
Limit concentration test only (LIMIT test). No mortality was observed.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)
Limit concentration test only (LIMIT test). No mortality was observed.

Irritation

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of irritating effects:
Not irritating to the skin. Not irritating to the eyes.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
Skin corrosion/irritation
rabbit: non-irritant (OECD Guideline 404)

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
Serious eye damage/irritation
rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of carcinogenicity:
No data was available concerning carcinogenic activity.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:
Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available data, the classification criteria are not met.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of repeated dose toxicity:
Repeated oral uptake of the substance did not cause substance-related effects.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Toxicity to fish:
LC50 (96 h) 10,250 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 84/449/EWG, C.1, static)
The statement of the toxic effect relates to the analytically determined concentration.

LC50 (96 h) > 1,004 mg/l, Oncorhynchus mykiss (OECD Guideline 203, semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna (Directive 79/831/EEC, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Aquatic plants:
EC50 (72 h) > 1,059 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Microorganisms/Effect on activated sludge:
EC20 (30 min) > 1,000 mg/l, activated sludge, domestic (DIN EN ISO 8192)
Nominal concentration.

12.2. Persistence and degradability

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). Inherently biodegradable. The product is biodegradable after extended adaptation.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Elimination information:

< 10 % DOC reduction (28 d) (DIN EN ISO 7827) (municipal sewage treatment plant effluent)

> 90 % DOC reduction (28 d) (OECD Guideline 302 B) (activated sludge, domestic)

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of stability in water:
In contact with water the substance will hydrolyse slowly.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Information on Stability in Water (Hydrolysis):
$t_{1/2}$ > 365 d (pH 7, 25 °C, pH value 7), (calculated)

12.3. Bioaccumulative potential

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

12.4. Mobility in soil

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

12.6. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.7. Additional information

Other ecotoxicological advice:

The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The local regulations on waste-water treatment must be followed.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Incinerate in suitable incineration plant, observing local authority regulations.

The UK Environmental Protection (Duty of Care) Regulations (EP) and amendments should be noted (United Kingdom).

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

SECTION 14: Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport
ADN

Not classified as a dangerous good under transport regulations

UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)

See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no
Classification applies for standard conditions of temperature and pressure.

The data should be considered when making any assessment under the Control of Substances Hazardous to Health Regulations (COSHH), and related guidance, for example, 'COSHH Essentials' (United Kingdom).

15.2. Chemical Safety Assessment

Product is not classified as hazardous.

SECTION 16: Other Information

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code.

ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.